

Baseline real-time target machine

SN5630

Technical Reference Information and Full System Test

Prepared by Speedgoat for:

McGill University

Content

1	Technical specification	5
1.1	Simulink Real-Time I/O connectivity	5
1.1.1	Installed Speedgoat I/O modules	5
1.1.2	Ethernet controllers.....	5
1.1.3	Storage device	5
1.1.4	CPU	5
1.2	Real-time target machine (software).....	6
1.2.1	Operating System (OS).....	6
1.2.2	DOS Drivers	6
1.2.3	Installed DOS tools	6
1.2.4	Simulink Real-Time kernel	6
2	Test model.....	7
2.1	Terminal boards test wiring	7
2.2	Simulink test model and results	9
3	I/O driver block library	11

1 Technical specification

1.1 Simulink Real-Time I/O connectivity

1.1.1 Installed Speedgoat I/O modules

Module	PCI [Bus, Slot]	IRQ line	Additional info
IO614	[8,0]	5	
IO391	[12,0]	11	FPGA 50k
IO183(1)	[6,0]	10	
IO183(2)	[11,0]	10	

1.1.2 Ethernet controllers

Component	PCI [Bus, Slot]	IRQ line	Supported
1. Host link	[16,0]	2	Host-target communication (Host link)
2. Ethernet port ETH 2	[14,0]	5	Real-Time UDP, EtherCAT Master
3. Ethernet port ETH 1	[13,0]	10	Real-Time UDP, EtherCAT Master
4. On board	[5,0]	5	Not connected

1.1.3 Storage device

Main Drive SSD 32GB ExtTemp

1.1.4 CPU

Processor Intel(R) Celeron(R) CPU J1900 @ 1.99GHz

1.1.5 Networking

Ethernet port HOST LINK	Enabled	Intel I210	00-01-29-89-6A-05
Ethernet port ETH1	Enabled	Intel I210	00-01-29-93-AB-54
Ethernet port ETH2	Enabled	Intel I210	00-01-29-93-AB-55

For Host / Target communication use the Ethernet port labeled "Host link"

1.2 Real-time target machine (software)

1.2.1 Operating System (OS)

Free DOS	Version 1.0
----------	-------------

1.2.2 DOS Drivers

USB driver	DOSUSB 2.0
------------	------------

1.2.3 Installed DOS tools

Simulink Real-Time kernel USB transfer tool	prepared by Speedgoat
---	-----------------------

1.2.4 Simulink Real-Time kernel

By default Speedgoat already preinstalls the latest Simulink Real-Time kernel version on your target machine. However to work with Simulink Real-Time a valid Simulink Real-Time license from The MathWorks is required.

Further information on how to transfer Simulink Real-Time kernel version for the respective MathWorks software release to your target machine and to change IP settings of your target machine is available in the Quick Start Guide and the User's Manual provided together with your target machine.

2 Test model

To test the provided I/O connectivity but also as an example on how to configure the provided driver blocks each target machine delivery includes a test model exercising various channels and ports of the installed I/O modules.

The test model uses the loopback method: the model generates outputs for the output pins of the I/O modules which are then routed back to the input pins of the I/O modules via simple wires wired to the terminals of the provided terminal boards or using loopback test cables such as for the IO601 (CAN).

You can monitor the transferred data on scopes of type target either by attaching a screen to your target machine, or take a screenshot by typing `tg = slrt; viewTargetScreen(tg);` at the MATLAB command prompt.

Note

Prior to executing the test model please ensure that you have properly configured and setup your target machine as explained in the Quick Start Guide and/or in the User's Manual provided with your target machine.

2.1 Terminal boards test wiring

To allow you to run the provided test model wire the terminal-board(s) as described below.

IO614

Connect:

CAN port A with CAN port B

CAN port C with CAN port D

using the provided CAN loopback test cables.

IO391

Connector A

From Pin		To Pin	
3	DO channel 1	9	DI channel 7
4	DO channel 2	10	DI channel 8
5	DO channel 3	11	DI channel 9
6	DO channel 4	12	DI channel 10
7	DO channel 5	13	DI channel 11
8	DO channel 6	14	DI channel 12

Connector B

From Pin		To Pin	
3	DO channel 13	9	DI channel 19
4	DO channel 14	10	DI channel 20
5	DO channel 15	11	DI channel 21
6	DO channel 16	12	DI channel 22
7	DO channel 17	13	DI channel 23
8	DO channel 18	14	DI channel 24
7	DO channel 17	15	DI channel 25
8	DO channel 18	16	Interrupt Input

IO183**Connector A**

From Pin		To Pin	
1	Analog Output 01	7	Analog Input 01
2	Analog Output 02	8	Analog Input 02
3	Analog Output 03	9	Analog Input 03
4	Analog Output 04	10	Analog Input 04

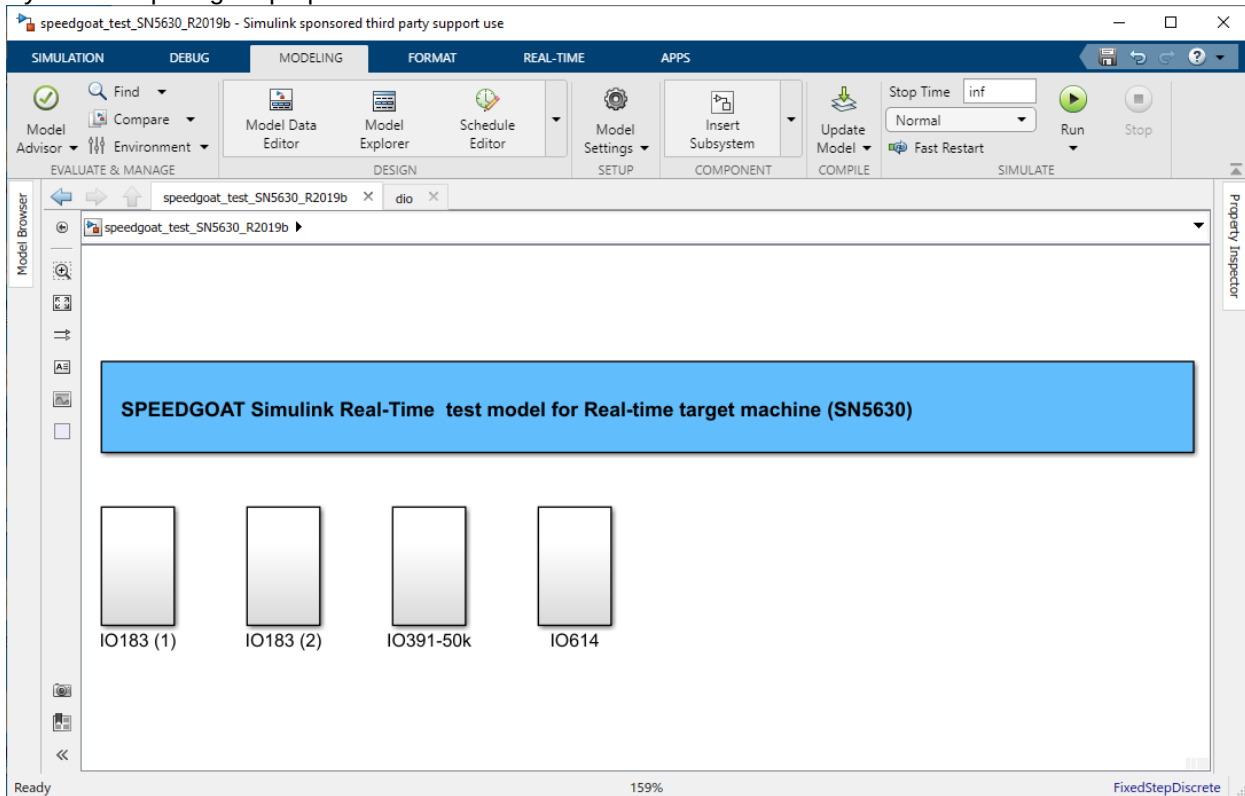
Connector B

From Pin		To Pin	
1	Digital I/O 01	9	Digital I/O 09
2	Digital I/O 02	10	Digital I/O 10
3	Digital I/O 03	11	Digital I/O 11
4	Digital I/O 04	12	Digital I/O 12
5	Digital I/O 05	13	Digital I/O 13
6	Digital I/O 06	14	Digital I/O 14
7	Digital I/O 07	15	Digital I/O 15
8	Digital I/O 08	16	Digital I/O 16

2.2 Simulink test model and results

As soon as you have connected the installed I/O modules either with the pre-wired terminal boards using the provided cables or with loopback test cables (such as with DSUB 9 connectors in case you have an IO601 installed or a crossed Ethernet cable in case you have an IO702 installed) you are ready to work with the provided test model(s).

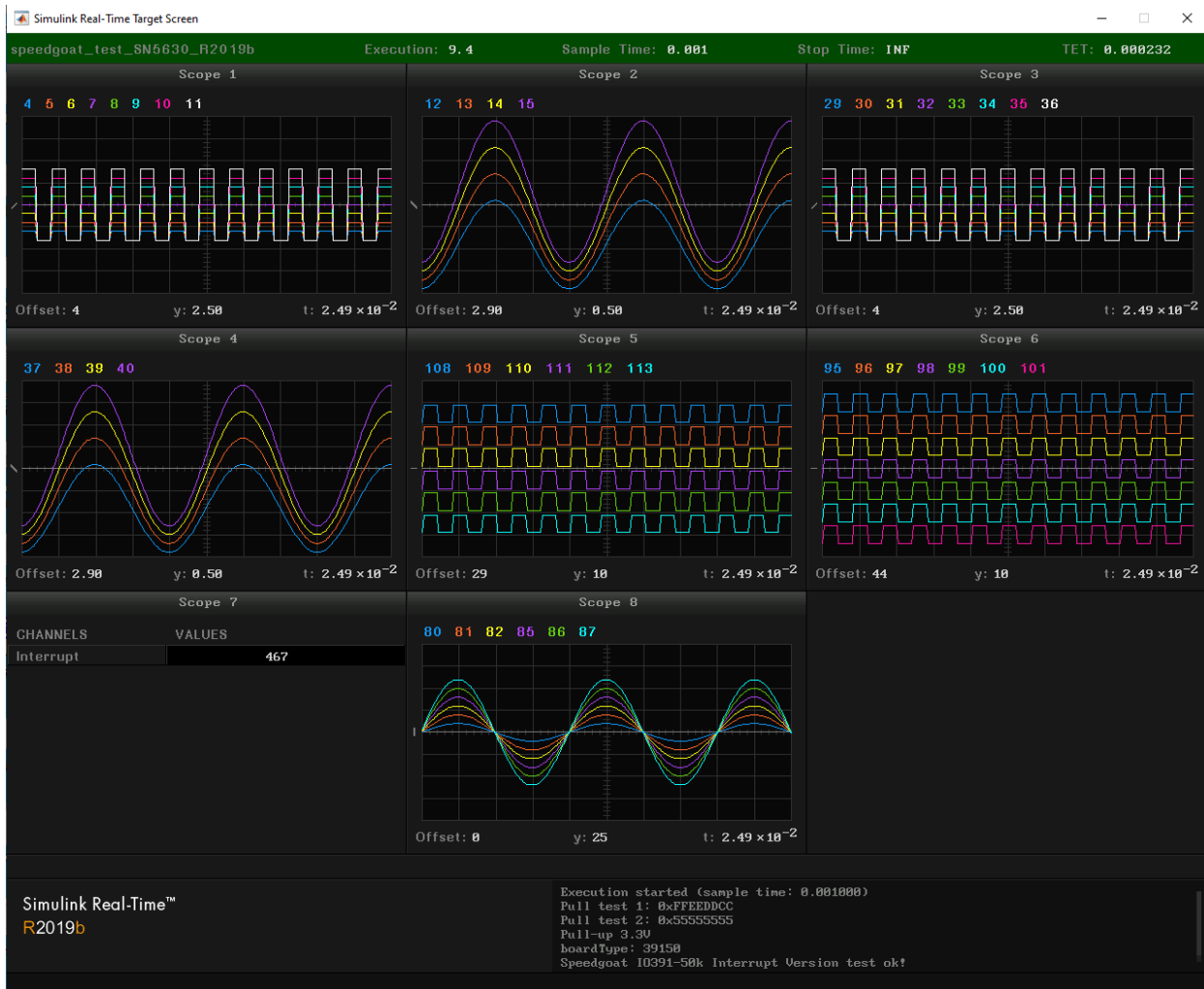
By default Speedgoat prepares test models for the current release of MathWorks software.



To get a better understanding about the structure of the test model you can open the top-level subsystems.

Before building the target application by selecting menu item "Tools/Code Generation/Build model" ("Tools/Real-TimeWorkshop/Build model" for R2010b and older) please ensure that you have defined a C-compiler compatible with Simulink Real-Time to compile and link the target application within the Simulink Real-Time Explorer, and that the Simulink Real-Time kernel has been selected and transferred to the target. For further information please refer to the Quick Start Guide and/or the User's Manual. Additionally, the target machine must be powered-up and the screen attached to the target machine must show the initialization screen of the loaded Simulink Real-Time kernel.

The target application generation process should take approximately 20-40 seconds. After the target application has been successfully downloaded, start the application by either typing `start(tg)` at the MATLAB command prompt or by using Simulink Real-Time Explorer. After a run-time of approximately 5 second the target scopes are updated for the first time and should look as follows.



If the target screen looks different please contact Speedgoat.

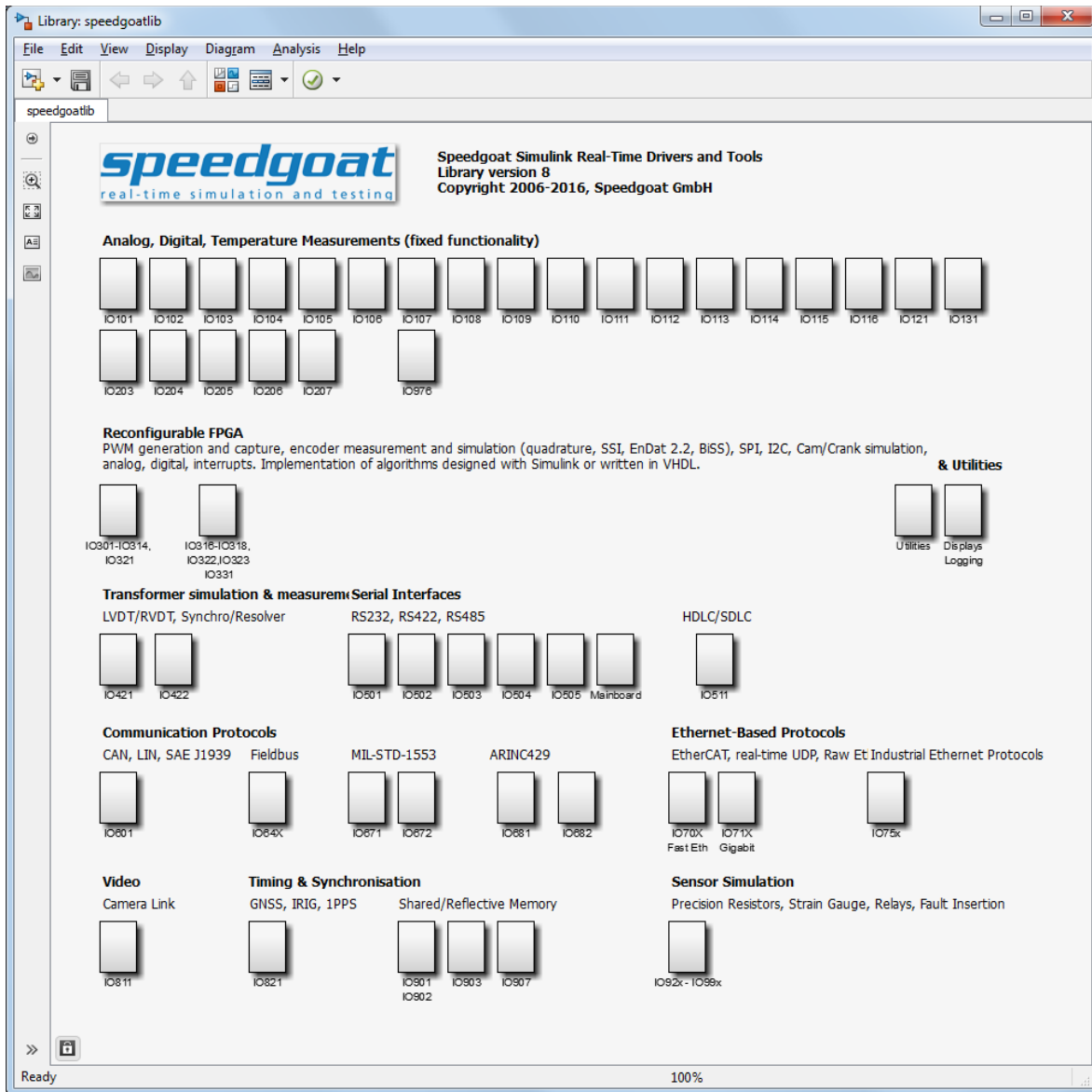
Note that the model is set to run infinite. To stop the model either type `stop(tg)` at the MATLAB command prompt or use the Simulink Real-Time Explorer.

Note

Besides testing the proper functioning of the installed I/O modules the test mode is also a good starting point for your own models. You can save it under a different name and then change whatever is needed.

3 I/O driver block library

Enter `speedgoatlib` at the MATLAB command prompt to open the Speedgoat driver block library.



It is important to not alter this library. It is a reference library. Drag and drop the driver blocks corresponding to the I/O modules installed in your target machine into your own Simulink models.

Information on how to configure and use the driver blocks is provided in the Speedgoat User's Manuals for the installed I/O modules.